A Generalization Sample Learning Method of Deep Learning for Semantic Segmentation of Remote Sensing Images

Chen Zheng, *Member, IEEE*, Jingying Li, *Graduate Student Member, IEEE*, Yuncheng Chen, *Graduate Student Member, IEEE*, and Leiguang Wang

***Abstract*— Deep learning methods have been widely studied in the semantic segmentation field of the remote sensing image. Training images play an important role in these methods; however, each training image usually contains not only the generalization information of each land category but also the specific interclass context between different categories. The specific interclass context prevents deep learning methods from focusing on generalization information learning during training and limits the performance on different data distributions. This article proposes a generalization sampling learning method of deep convolutional neural network (GSL-CNN) to emphasize generalization information learning for the semantic segmentation of remote sensing images. The proposed method develops a new CBR sampling strategy that contains three modules: category grouping (*C*), basic unit extraction (*B*), and random combination (*R*). Module *C* collects each land category map and strips away the specific interclass context from the raw annotated image. Module *B* extracts basic units with different granularities from each land category map, and each basic unit can keep the generalization information of this category. Module *R* aims to enhance the robustness against different data distributions by randomly picking basic units of different categories and randomly generating their interclass context. The new GSL-CNN method integrates the CBR sampling strategy with the convolutional neural network (CNN) model for semantic segmentation. Experiments on different remote sensing datasets and 15 state-of-the-art CNN models validated that the proposed method has the potential of improving the generalization ability of the CNN method from a sampling perspective.**

***Index Terms*— Deep learning, generalization, remote sensing, sampling, semantic segmentation.**

Manuscript received 7 January 2023; revised 10 April 2023 and 16 June 2023; accepted 22 July 2023. Date of publication 26 July 2023; date of current version 9 August 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 41771375, Grant 31860182, and Grant 41961053; in part by the Natural Science Foundation of Henan under Grant 232300421071; in part by the Scientific and Technological Innovation Talent in Universities of Henan Province under Grant 22HASTIT015; and in part by the Youth Key Teacher of Henan under Grant 2020GGJS030. *(Corresponding author: Jingying Li.)*

Chen Zheng is with the School of Mathematics and Statistics and the Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, Institute of Applied Mathematics, Henan University, Kaifeng 475004, China (e-mail: zhengchen_data@126.com).

Jingying Li and Yuncheng Chen are with the School of Mathematics and Statistics, Henan University, Kaifeng 475004, China (e-mail: 104752210083@henu.edu.cn; iichengzi@henu.edu.cn).

Leiguang Wang is with the Institute of Big Data and Artificial Intelligence, Southwest Forestry University, Kunming 650224, China (e-mail: wlgbain@126.com).

Digital Object Identifier 10.1109/TGRS.2023.3298924

I. Introduction

SEMANTIC segmentation is one fundamental and important task in the field of remote sensing image processing. It aims to assign each pixel of an image a land cover or land use type. Semantic segmentation has been widely used in various fields such as urban planning, precision agriculture, and environmental assessment. With the development of machine learning, deep learning-based semantic segmentation has become the most popular and effective idea and has achieved a great progress in practice [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. However, the performance of deep learning methods depends on the quantity and quality of training images.

Recently, many methods have been proposed to enhance the generalization ability of deep learning. These methods can be roughly divided into two groups: interextension and intraextension. The former focuses on how to introduce more raw samples at a low cost, instead of pixel-wise annotated labeled images [11], [12], [13], [14], [15], [16]. Semi-supervised learning and weakly supervised learning are two main methods. The semi-supervised learning uses unlabeled samples to expand training samples to improve performance [17], [18]. The weakly supervised learning uses the sparse pixel-wise labels or the image-wise labels [19] to achieve the semantic segmentation. These methods can effectively enhance the generalization ability by taking more training information into account, but there are a lot of mislabeled samples in the interextension data [20]. The intraextension, on the other hand, puts attention on data augmentation of existing annotated samples. It can be further classified into two groups. One group employs the deep convolutional neural networks (CNNs) for data augmentation. In particular, the generative adversarial network (GAN) has attracted much attention as it can generate synthetic samples by using the limited annotated samples to augment the training samples [21], [22], [23], [24], [25], [26]. The other group uses data augmentation tricks to increase the number of training images. Specifically, some data augmentation tricks generate new training image by making geometric transformation to the original training images, such as flipping [27], cropping [28], and rotating [29]. These tricks can improve the generalization ability against the orientation and location. In addition, regional dropout

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

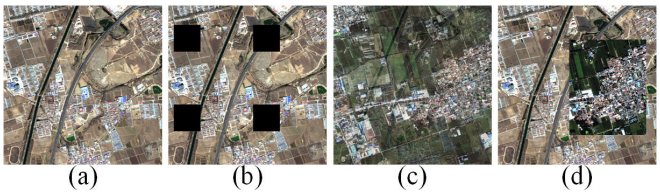


Fig. 1. Examples of (a) random crop, (b) Gridmask, (c) Mixup, and (d) Cutmix in the remote sensing images.

strategies are proposed to enhance the generalization ability of CNNs by randomly removing some regional patches from the training image, such as Gridmask [30] shown in Fig. 1(b). This technique of changing contextual information can help the model to achieve better performance against various contexts, but it reduces the proportion of informative pixels on the training image. Mixup [31] generates new training images via convex combinations of pairs of training images and their labels to reduce overfitting. However, as shown in Fig. 1(c), Mixup example confuses the feature space and spatial relationship information of the training data, which may mislead the learning of the model. Cutmix [32] overcomes the problem of Gridmask and Mixup by filling the removed regions with a patch from another training image, as shown in Fig. 1(c). This method can provide better results on natural images by effectively utilizing training pixels and retaining the advantages of region dropout. The above works indicate that the learning of contextual information can help the model improve generalization performance.

For the contextual information, it has two different types. One is the intraclass context and the other is the interclass context. Although different types of images usually contain these two context information, the remote sensing image has a different representation of these contexts than other images such as natural images. Specifically, the interclass context of remote sensing images is more random compared to natural images. That is to say, the intraclass and interclass contexts play different roles in the remote sensing images. However, the above-mentioned data augmentation methods treat them as a whole.

To address this problem, exploring the mechanism of generalization ability for remote sensing images is required. Fig. 2 shows the process of training image generation. Such a process is usually randomly sampling with a certain size from the raw annotated remote sensing image. For these training images, they contain two types of information. One is the individual information of each land category, which includes both the feature distribution of this category and the intraclass context. They are the generalized concepts that could be used for various types of remote sensing images. The other is contextual information between different land categories, i.e., the interclass context. It is usually a specific information that can only fit some certain data distributions [33]. Hence, to improve the generalization ability of deep learning for remote sensing images, more attention should be paid to learning the generalized homogeneous information from objects that have various appearances but belong to the same land category, as well as the intraclass context. At the same time, the specific interclass context should be suppressed

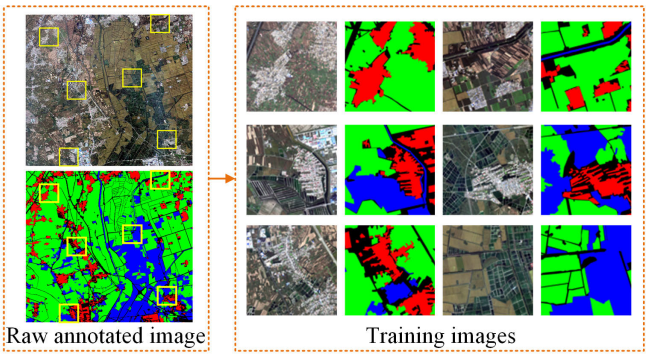


Fig. 2. Visualization of training images generation.

during training, otherwise this will disperse CNNs to learn the generalization concept. However, current data augmentation methods usually directly use the patches cropped from the raw remote sensing images or their combination to generate the training image. These methods mix the above two types of information from the raw annotated image.

To highlight the learning of the generalization information for each individual land category, this article proposes a novel generalization sampling learning method of deep CNN (GSL-CNN) for semantic segmentation of remote sensing images. The GSL-CNN develops a new three-module sampling strategy, i.e., "Category grouping"-"Basic unit extraction"-"Random combination" (CBR), to emphasize the learning of generalization information for remote sensing images. Specifically, in the module of category grouping, the GSL-CNN extracts each category map from the raw annotated remote sensing image. For each category map, it can strip away the inherent pattern of the interclass context and focus on learning the generalization information of current land category. All category maps constitute the category grouping, as illustrated in Fig. 3 (module $\boldsymbol{C}$). Then, the basic unit extraction module uses an adaptive local window to capture the basic units of each category map. Note that each basic unit only contains one category information. It can represent the concept of each category from different granular parts, and consider the characteristics of each category from the micro to the macro, such as the intraclass spatial contextual relationship, shape, and intraclass texture, as illustrated in Fig. 3 (module $\boldsymbol{B}$). Modules $\boldsymbol{C}$ and $\boldsymbol{B}$ are developed to capture the generalization information within an individual category. In the last random combination module, the training images are generated by randomly integrating different basic units of all the categories, which is possible to break through the limitations of the interclass context from raw images and enhance the effectiveness of the GSL-CNN method against different data distributions. The main contributions of this article lie in the following three aspects.

1) The proposed method divides the raw image into different category maps and generates the training image according to the basic units clipped from different category maps, where each basic unit, i.e., the patch that constitutes the training image, only contains one category information to emphasize the generalization information within an individual category. It is the critical difference from current data augmentation methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

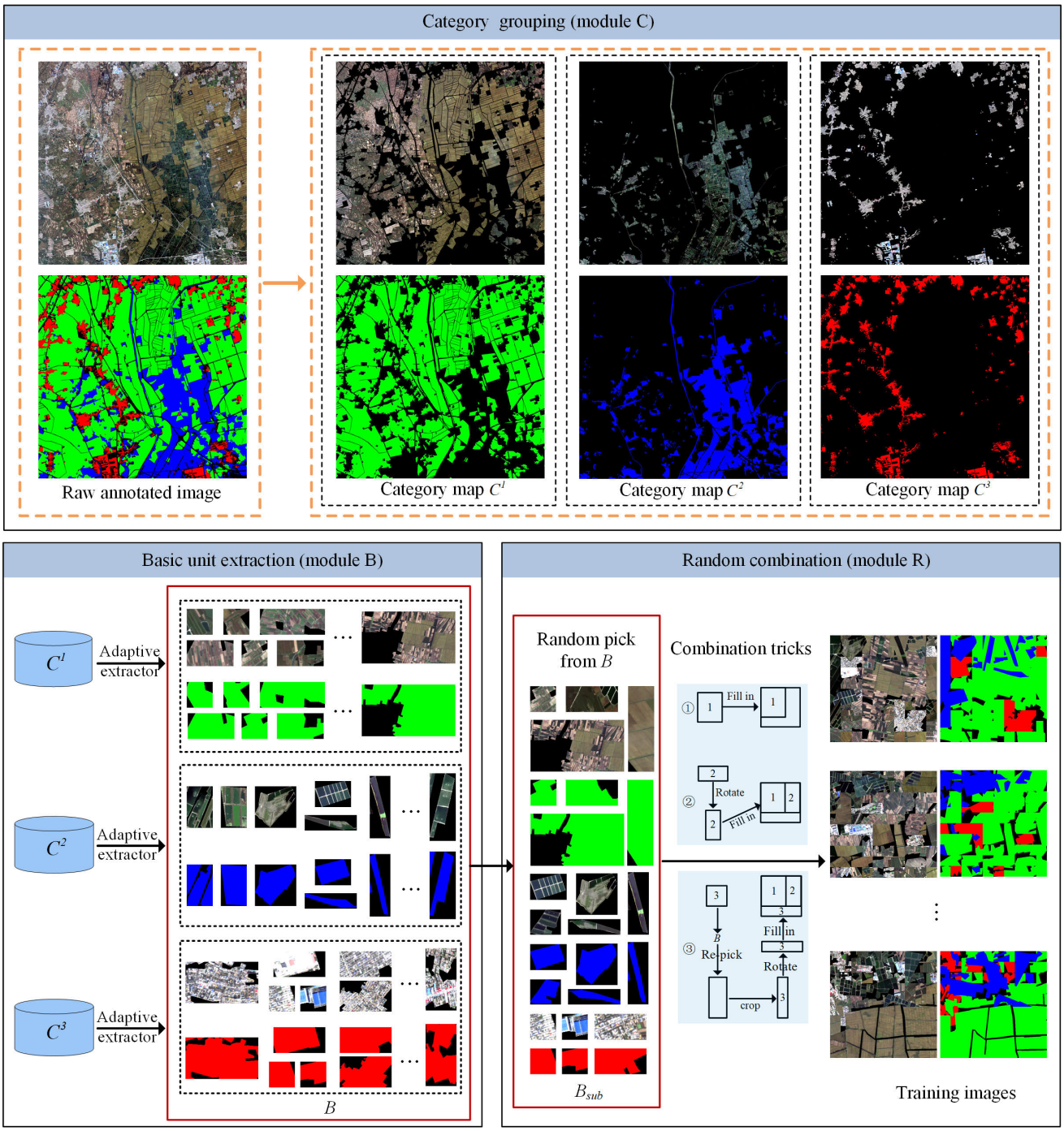


Fig. 3. Framework of the proposed CBR strategy, which is composed of three modules, namely category grouping (module ***C***), basic unit extraction (module ***B***), and random combination (module ***R***). Module ***C*** extracts each individual land category map from the raw annotated image. Module ***B*** uses an adaptive extractor to collect basic units that capture the category generalization information. Module ***R*** generates training images by random pick and combination tricks.

2) A new sampling strategy, CBR, is developed to enhance the generalization ability of CNNs by emphasizing the generalized learning of each category concept during the training process.
3) A semantic segmentation framework, the GSL-CNN method, is proposed to provide segmentation result by integrating the CBR strategy into the CNN methods.

II. Proposed Method

The proposed GSL-CNN method aims to emphasize the learning of generalization information during the training process. This section first reviews the classic sampling crop method and data augmentation methods for enhancing generalization ability, then discusses the CBR strategy of the proposed method in detail, and finally introduces the GSL-CNN method.

A. Sampling Strategy of the Crop

Let $I = \{I_s | s \in S\}$ and $L = \{L_s | s \in S\}$ denote the raw observed remote sensing image and its pixel-wise annotated image, respectively. Here, $S = \{s = (x, y) | 1 \le x \le m, 1 \le y \le n\}$ is the index set and each s in S denotes the site of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

each pixel. Each L_s of L represents the land category for site s and takes value from the set $\Lambda = \{1, 2, \ldots, k\}$, where k is the number of land categories.

If T_i, sized $m_1 \times n_1$, denotes a local block of the observed image I, such as the area marked with the box in Fig. 2. The sampling strategy of the crop is usually to divide the observed image I into $((m - m_1/\tau) + 1) \times ((n - n_1/\tau) + 1)$ different local blocks by using a sliding window with step length τ. And then it will randomly select q local blocks and form a training image collection $T = \{T_i | i = 1, 2, \ldots, q\}$, where each T_i is treated as the training image and $q \leq ((m-m_1/\tau)+1) \times ((n-n_1/\tau)+1)$. Some works will require that $T_i \cap T_j = \emptyset$ for $i \neq j$ and further implement rotating or geometric transformation on T [19], [34] to improve the generalization ability.

The above crop method and its variations use random sampling strategy and related tricks to prevent overfitting, but the interclass context of the selected training image is relative stable. It will not only cost more training images, but also perform worse if test images are out of the learned specific data distribution [35]. To further improve the training performance, several generalization tricks, such as Mixup [31], Cutmix [32], and Gridmask [30], are proposed for training data augmentation. Mixup provides smoother uncertainty estimates for decision boundaries and reduces overfitting to the training data. Cutmix and Gridmask help a model focus more on objects and the most discrimination parts of object regions by dropping or mixing a random region in the training image.

Above methods show that data augmentation tricks such as rotating, geometric transformation, regional dropout, or mixup method can enhance the generalization ability of the CNN by preprocessing the training images. Moreover, regional dropout strategy, such as Gridmask, indicates that the generalization concept of one land category can be effectively learned within a certain block, even its neighboring regions are dropped. Hence, enhancing the generalization ability should focus more on the intraclass relationship within each individual land category. Meanwhile, mixup method, such as Cutmix and its application in the remote sensing image processing [36], indicates that reorganizing the context relationship among different land categories would not hinder the generalization learning.

B. CBR Sampling Strategy

To explore the mechanism of enhancing the generalization ability when training a CNN model, a new generalization sampling strategy inspired by the theoretical knowledge of stratified sampling [37], the CBR sampling strategy, is proposed in this part. According to the analysis of the above methods, CBR sampling strategy is mainly developed to emphasize the learning of generalization information of each land category and break the limitation of the interclass context among different land categories.

An overview of the CBR sampling strategy is illustrated in Fig. 3, which contains three modules: category grouping ($\boldsymbol{C}$), basic unit extraction ($\boldsymbol{B}$), and random combination ($\boldsymbol{R}$). The pipeline of the CBR sampling strategy has three steps: 1) collecting each land category map from the raw annotated image L in module $\boldsymbol{C}$; 2) extracting basic units with different sizes from category maps; and 3) combining various basic units into a training image sized $m_1 \times n_1$. Here, modules $\boldsymbol{C}$ and $\boldsymbol{B}$ are designed to highlight the generalization concept of each category, and module $\boldsymbol{R}$ generates the training image with a random interclass context between basic units of different categories. The details of the proposed modules $\boldsymbol{C}$, $\boldsymbol{B}$, and $\boldsymbol{R}$ are discussed in the following.

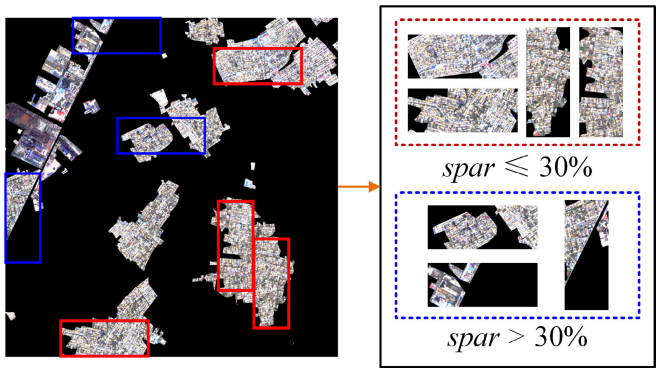


Fig. 4. Illustration of same-size basic unit extraction in module $\boldsymbol{B}$. Basic units of the built-up, sized $m' \times n'$, are extracted in this figure and the sparsity is designed to distinguish whether a basic unit captures enough generalization information (sparsity $\leq$ 30%, red box) or not (sparsity $>$ 30%, blue box).

1) Module C: This module proposes the category map $C^l = \{C_s^l | s \in S\}$ to extract the information of the category l from the raw observed image I, $l \in \Lambda = \{1, 2, \ldots, k\}$. In C^l, each C_s^l is defined as

$$C_s^l = \begin{cases} I_s, & \text{if } L_s = l \\ 0, & \text{if } L_s \neq l \end{cases} \tag{1}$$

as shown in Fig. 3. Each category map C^l can focus on the representation of characteristics and the intraclass context within the l th category, and the collection of all category maps constitutes the category grouping $\boldsymbol{C} = \{C^1, C^2, \ldots, C^k\}$.

2) Module B: In this module, each category map C^l in the category grouping $\boldsymbol{C}$ will be clipped into different basic units to generate the training images. The core issue of module $\boldsymbol{B}$ is that each basic unit should keep the generalization information of the corresponding category during clipping. It has two key factors: the size and sparsity for each basic unit.

For the first factor, setting the size of the basic unit, it is guided by the idea that the percentage of each category in the training image should remain the same as the observed image. Specifically, if the percentage of the category l in the observed image is P^l, i.e., $P^l = \text{num}\{s | L_s = l, s \in S\}/(m \times n)$, then, the size of the basic unit for category l, $m' \times n'$, should satisfy that

$$\frac{m' \times n'}{m_1 \times n_1} = P^l \tag{2}$$

where $m_1 \times n_1$ is the size of each training image and $\text{num}\{s | L_s = l, s \in S\}$ denotes the number of pixels in the set $\{s | L_s = l, s \in S\}$. In this module, different size combinations $m' \times n'$ satisfying (2) are all available, as shown in Fig. 4.

However, when sliding the window to extract basic units that meet the size condition of (2), due to the sparsity of each category map, some basic units may only contain a small number of nonzeros observed points, as illustrated in the blue box in Fig. 4. This will lead to lower learning efficiency for the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

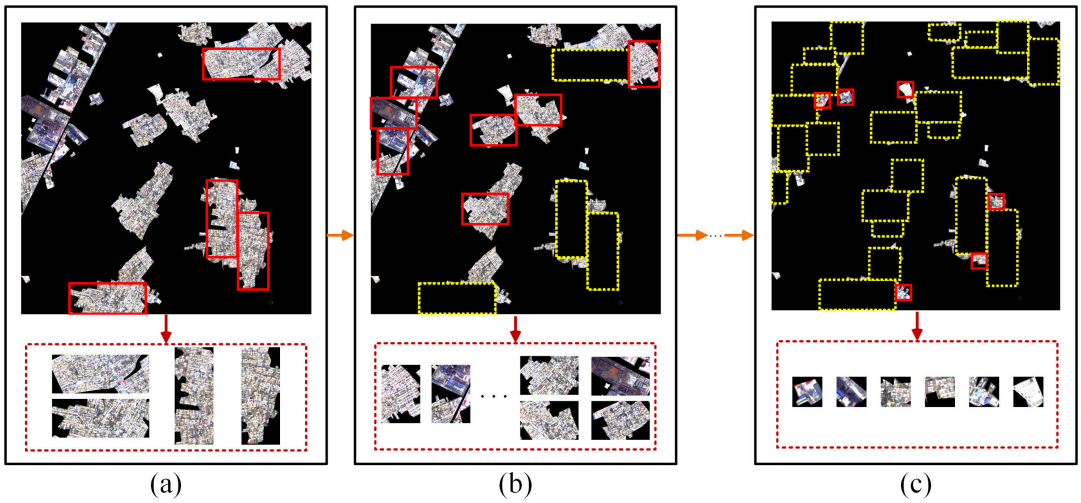


Fig. 5. Illustration of decreasing-size basic unit extraction in module $\boldsymbol{B}$. Under the same size, basic units satisfying spar $\leq 30\%$ are collected from the category map, as shown in (a). Otherwise, the size is reduced and a sequence of basic units is collected with different sizes, as shown in (b) and (c).

proposed method. To alleviate this problem, the second factor of basic unit extraction is developed to control the sparsity of each basic unit. If B_i^l denotes one basic unit of the category l, $i = 1, 2, \ldots, N_l$, the sparsity of B_i^l is defined as

$$\text{spar} = \frac{\text{num}\{s | C_s^l = 0, s \in B_i^l\}}{m' \times n'}. \tag{3}$$

Module $\boldsymbol{B}$ of the CBR sampling strategy only considers the basic units with spar $\leq 30\%$.

Under the constraints of above two factors, only densely distributed areas will be clipped as the basic units, as shown in the red box in Fig. 4. In this way, the information of sparse distributed areas will be lost, as shown in the blue box in Fig. 4. To fully utilize the information of the raw annotated image, a sequence of basic units with the decreasing size is further developed, which keeps spar $\leq 30\%$ but relaxes the constraint of the size factor, as shown in Fig. 5. This makes module $\boldsymbol{B}$ can collect densely distributed areas with different granularities. It leads these basic units can capture the characteristics of each category from different perspectives and enhance the ability of the generalization information learning.

3) Module R: This module is used to randomly pick basic units of different categories and combine them as the training image R sized $m_1 \times n_1$. When randomly picking the basic unit(s) for each category l, every basic unit B_j^l has the same chance to be picked, and the total size of the selected basic unit(s) satisfies the condition in (2), that is,

$$\sum_j \text{size}\left(B_j^l\right) = P^l \times (m_1 \times n_1). \tag{4}$$

Picked basic units of all the categories form a collection B_{sub}, as illustrated in Fig. 6(I).

This module further proposes the combination tricks to generate the training image R according to the collection B_{sub}. During the process of combination, R is first initialized as an empty image of size $m_1 \times n_1$. Then, the current largest basic unit B_j^l in B_{sub} is selected each time, which will be removed from B_{sub} at the same time. Let B_j^l fill in an empty block of R, as shown in Fig. 6(i), (ii), and (v). If B_j^l cannot fit the empty block, but B_j^l rotated by 90° can fit the empty block. Then, rotate B_j^l and fill it to R, as shown in Fig. 6(iii). If the size of some basic units prevents them from being filled into R, no matter rotating or not, this module will randomly repick the basic unit of corresponding category and crop it to the appropriate size to fill in R until there are no empty blocks in R, as shown in Fig. 6(iv). When all the basic units in B_{sub} are filled to R, the training image R with a random the interclass context can be generated based on their combination.

C. Proposed the GSL-CNN Method

According to the CBR sampling strategy, this section proposes the GSL-CNN method to provide a framework to improve the generalization ability of the CNN-based methods. Specifically, the GSL-CNN method consists of two modules. The first module, the CBR sampling strategy, is used to generate training images from the raw annotated images, and then trains a certain CNN-based method based on these training images at the second module. The workflow of the GSL-CNN method is visually shown in Fig. 7.

III. Experiments

The proposed GSL-CNN method provides a CBR sampling strategy to emphasize the representation of the generalization information when generating training images from raw annotated images. It aims to establish a general framework to improve the generalization ability of the CNN-based methods from the sampling perspective. To experimentally test the effectiveness of the proposed GSL-CNN method, different types of CNN-based methods with the CBR sampling strategy are evaluated over various types of remote sensing images and compared with other state-of-the-art data augmentation methods. This section includes three modules. First, the tested data and implementation details are given. Second, parameters setting are experimentally discussed for the CBR strategy of the GSL-CNN method. At last, to test the robustness against different CNN-based methods, performances of the GSL-CNN method based on 15 different state-of-the-art CNN methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

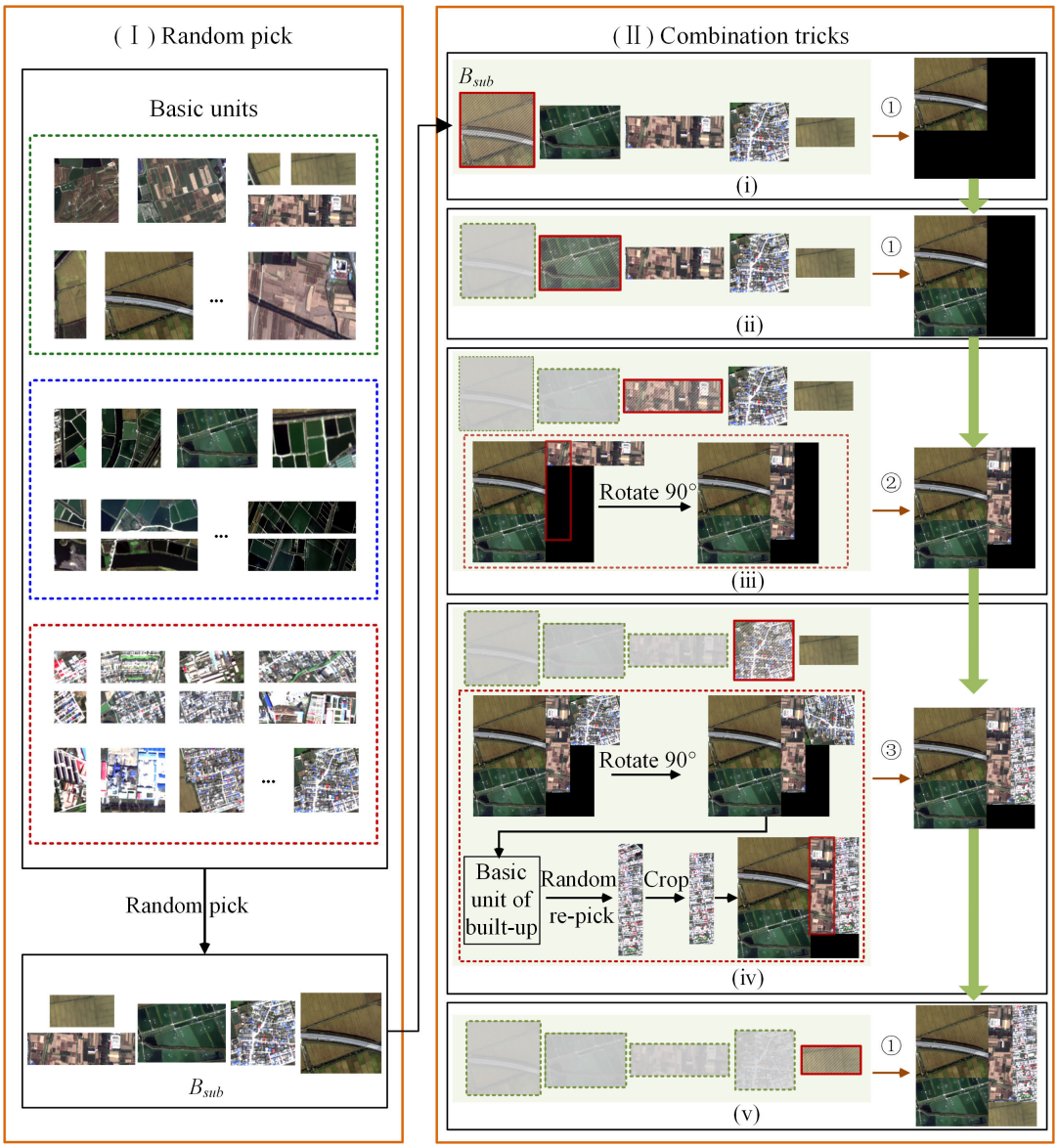


Fig. 6. Demonstration of the random pick and combination trick in module $\boldsymbol{R}$, where the training image will be generated by randomly picking basic units and integrating them with the combination tricks. Specifically, picked basic units in the B_{sub} will be combined to generate the training image through three combination tricks, that is: ① fill in [Steps (i), (ii), and (v)]; ② rotate 90° [Step (iii)]; and ③ re-pick and crop [Step (iv)]. (I) Random pick. (II) Combination tricks.

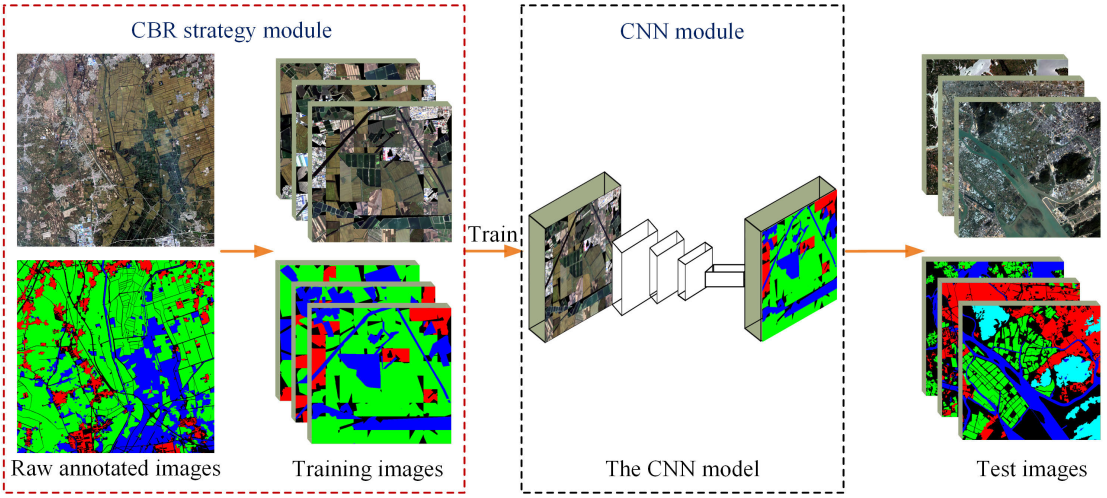


Fig. 7. Workflow of the GSL-CNN method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISON WITH DIFFERENT STEP SIZES (UNIT: %)

p value	F1-score					OA	Kappa
	Water	Built-up	Farmland	Forest	Meadow		
20	64.3	95.5	94.4	96.2	93.6	93.9	86.2
30	67.2	95.5	94.8	96.2	93.2	94.5	86.7
40	63.4	93.6	94.4	97.9	83.5	93.6	86.7
50	65.9	95.3	96.0	96.5	91.3	95.7	88.0
60	60.2	95.6	95.7	96.6	90.6	95.3	86.9
70	62.6	95.7	95.2	96.1	93.9	95.0	88.1
80	66.0	94.7	95.0	96.2	94.5	94.5	87.7
90	66.0	95.6	94.8	96.1	93.1	94.4	86.4

are tested on different datasets, which are also compared with other data augmentation methods.

A. Datasets and Implementation Details

1) Datasets:

a) Gaofen image dataset (GID): GID is a high spatial resolution remote sensing image dataset that consists of 150 Gaofen-2 (GF-2) satellite images [38]. These images cover more than 60 different cities in China, and the covered geographic areas is more than 50 000 km^2. Each image in GID sizes 7200 $\times$ 6800 with a 4 m spatial resolution. GID has annotated five categories, i.e., water, built-up, farmland, forest, and meadow. In this experience, 120 raw annotated images in GID are used for the training process, the rest are used for testing. Due to the large size of the GID images, the size of the training image generated from the annotated image is set to 600 $\times$ 600 in the experiment.

b) Sentinel-2 dataset: Sentinel-2 is a high-resolution and multispectral satellite manufactured by the European Space Agency. In this experiment, 40 cloud-free Sentinel-2 images, sized 1250 $\times$ 1250, are collected from the Google Earth Engineer [39] as the Sentinel-2 dataset in this article. The images of this dataset are of RGB bands with 10-m spatial resolution, obtained from 2020 to 2021. Four land categories are pixel-level annotated for this dataset according to the Land Use Classification Criteria of GB/T21010-2017, which are built-up, farmland, forest, and water. Thirty raw annotated images sized 1250 $\times$ 1250 are used for the training process and the rest ten images are used for testing. The size of each training image generated from the annotated image is set to 180 $\times$ 180 in the experiment.

2) Experimental Setting: The deep learning method used in the experiment is implemented based on the Pytorch framework. The optimizer is set to Adam [40] and the base learning rate is set to 0.01. A poly learning rate policy is employed in the training process, where the learning rate is multiplied by $(1 - (\text{iter}/\text{num_iter}))^{0.9}$ after each iteration [41]. The weight decay value is $5e^{-4}$. The training epoch is set to 50 for both GID dataset and Sentinel-2 dataset. The batch size is, respectively, set to 4 for GID dataset and 12 for Sentinel-2 dataset due to their different sizes of training images. In the experiments of each dataset, 3000 training images are generated from the annotated image for each comparison method and the proposed method. To quantitatively evaluate the performance of the proposed method on different datasets, three metrics are used to measure experimental results, i.e., $F1$-score, overall accuracy (OA), average accuracy (AA) [42], and Kappa [43].

B. Test of the CBR Strategy

1) Test of the CBR Strategy: In the CBR strategy, basic units in each land category aim to highlight the generalization concept of this individual category while limiting the impact of the interclass context as much as possible. In module $\boldsymbol{B}$ of the CBR strategy, when extracting a sequence of basic units with decreasing size, a key factor is how to determine its step size for each decreasement. Let p denote the step size of the decreasing size. It will not only bring a high computational cost but also fail to capture the generalization information when the size of the basic unit decreases to a too small value. While, on the other hand, a too big step size may lose some important information about land categories. Hence, the CBR strategy against different step sizes is experimentally tested on the GID images, and its quantitative indicators are shown in Table I.

As we can observe, in the interval $[20, 90]$ with stride 10, the proposed method will have a worse performance if the step size p is more than 70. At the same time, if p is less than 50, the quantitative indicators are also relative low as too small-sized basic units will be collected, which will lose the generalization characteristics of each land category. And the appropriate interval is $[50, 70]$ for the step size p and it is set to 60 in this article for all the GID experiments.

Moreover, from the $F1$-score of different land categories in Table I, it can be seen that water performs poorly compared to other land categories. It is due to that the spatial distribution of water is relatively sparse. In fact, when basic units of water require to meet the sparsity of module $\boldsymbol{B}$, i.e., (3), their size are usually quite small as the sparse spatial distribution of water, as shown in Fig. 8(b). It will cause that basic units would fail to fully capture the macro morphological structure of water. To alleviate this issue, a water-extraction trick is designed to learn the concept of water during training. Namely, in order to catch the macrospatial distribution of water, the size and sparsity of each basic unit break the limitation of (2) and (3) for water and are only determined according

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

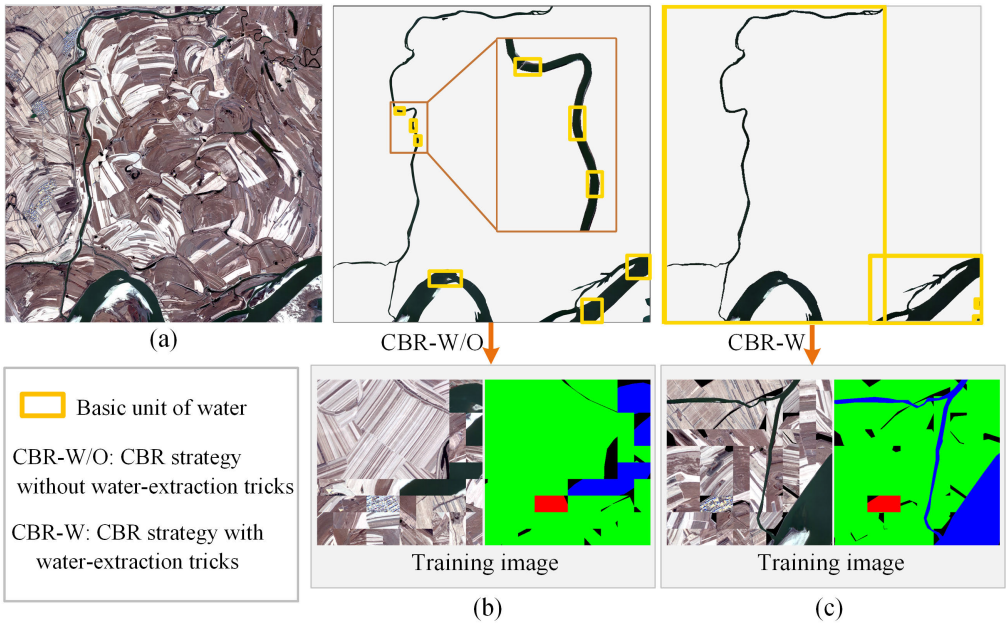


Fig. 8. Illustration of the basic unit extraction for water for CBR strategies with and without water-extraction tricks. (a) Raw annotated images. (b) CBR strategy without water-extraction tricks to generate training images. (c) CBR strategy with water-extraction tricks to generate training images.

TABLE II

QUANTITATIVE COMPARISON OF CBR STRATEGIES WITH AND WITHOUT WATER-EXTRACTION TRICKS, WHERE THESE TWO METHODS ARE ABBREVIATED AS CBR-W AND CBR-W/O, RESPECTIVELY (UNIT: %)

Method	F1-score	OA	Mean
	Water		F1-score
CBR-W/O	43.5	94.7	86.1
CBR-W	82.3	96.8	90.7

to the shape of each water. That is to say, the entire structure of each water will be extracted as a basic unit of water from the raw annotated image, as illustrated in Fig. 8(c). Note that the basic unit of water is too large to generate a training image, this trick will clip the water unit along the direction of water flow to keep its main spatial structure when fitting the proportion of each land category in the training image. Strategies with and without water-extraction tricks are evaluated and their numerical results are listed in Table II. It can be seen that numerical results with water-extraction tricks perform better than those without water-extraction tricks, where the $F1$-score can be improved by 39.8% for water, and the OA and Mean $F1$-score can be improved by 2.1% and 4.6% in total, respectively. It demonstrates that the CBR strategy with water trick can capture the concept of water more effectively. More experimental results with and without water tricks are demonstrated in Fig. 9. These results further validate the effectiveness of this trick, especially for the recognition of small tributaries of the water.

2) Test of Module ***C****:* Module ***C*** divides the annotated image into different individual category maps, which aims to focus on the learning of intraclass interaction and weaken the influence of specific interclass spatial distribution, as shown in Fig. 3. To test the role of module ***C***, the training image

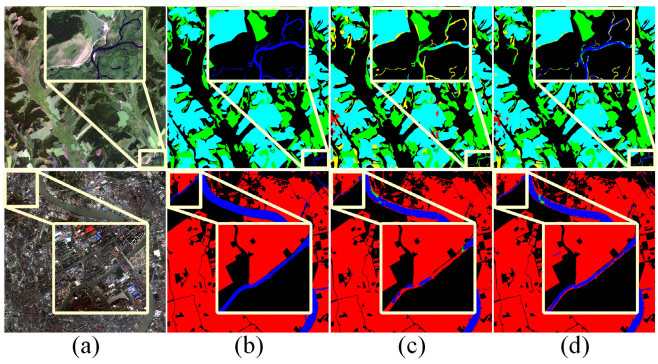


Fig. 9. Semantic segmentation results of CBR strategies with and without water tricks. (a) Original. (b) Ground truth. (c) CBR-W/O. (d) CBR-W.

that directly crops from the raw annotated image are used for comparison as it contains not only the intraclass interaction but also the inherent pattern of the interclass context, such as the instance shown in Fig. 2. Experimental results of these two methods are illustrated in Fig. 10, whose quantitative indicators are listed in Table III. As we can observe, module ***C*** can provide better performance and improve the generalization by using individual category maps, especially in the urban and water areas, which verifies the effectiveness of module ***C***.

3) Test of Module ***B****:* Module ***B*** aims to clip basic units from each category map, where each basic unit can keep the generalization information of this category. To address this issue, the decreasing-size and sparsity are used in module ***B*** as two key constraint factors. These factors can generate densely distributed basic units to capture the generalization information of each category more effectively. To verify the role of module ***B***, basic units without these two constraint factors are generated for comparison. Experimental results and their numerical indicators are presented in Fig. 11 and Table IV. It can be seen that basic units generated from module ***B***

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

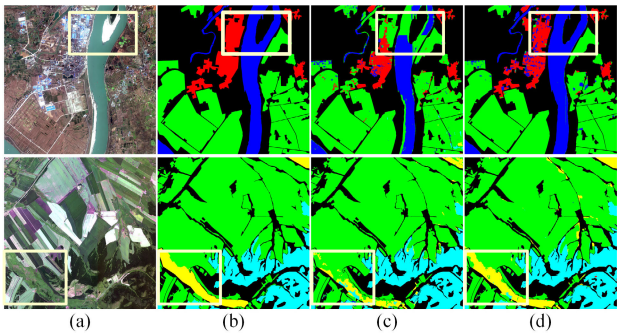


Fig. 10. Semantic segmentation results of method with specific data distribution and module $\boldsymbol{C}$ without specific data distribution. (a) Original remote sensing images. (b) Ground truth. (c) Results of method with specific data distribution. (d) Results of module $\boldsymbol{C}$ without specific data distribution.

TABLE III

QUANTITATIVE COMPARISON OF METHOD WITH SPECIFIC DATA DISTRIBUTION AND MODULE $\boldsymbol{C}$ WITHOUT SPECIFIC DATA DISTRIBUTION, WHERE THEY ARE ABBREVIATED AS $\boldsymbol{C}$-W AND $\boldsymbol{C}$-W/O, RESPECTIVELY (UNIT: %)

Indexes	$\boldsymbol{C}$-W	$\boldsymbol{C}$-W/O
Water	71.3	98.9
Built-up	58.2	83.7
Farmland	97.3	98.6
Forest	97.2	98.4
Meadow	59.6	85.7
OA	90.6	97.1
Kappa	82.4	94.1

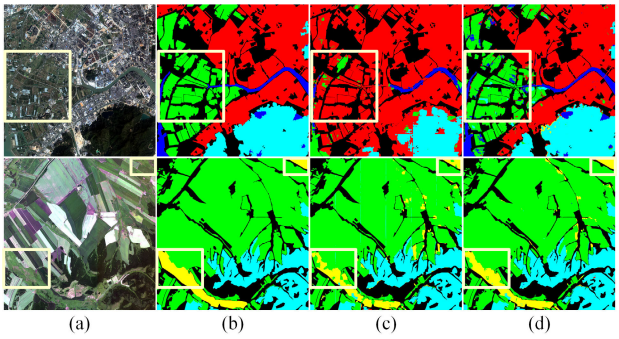


Fig. 11. Semantic segmentation results that extract basic units with and without two constraint factors in module $\boldsymbol{B}$. (a) Original remote sensing images. (b) Ground truth. (c) Results that extract basic units without two constraint factors. (d) Results of module $\boldsymbol{B}$.

consistently outperform those of comparison method, which improves the $F1$-score by 28.7% at the highest and 4.6% at least. These results validate the effectiveness of basic unit extraction in module $\boldsymbol{B}$.

4) Depth of the Model Backbone: This article selects ResNet as the backbone network for CNN-based methods. The depth of the model is an important factor that affects the size of the receptive field and also affects the extraction effect of the spatial relationship. Hence, ResNet with a depth of 18, 50, and 101 are, respectively, employed to test the performance of the proposed method against different depths. An experiment taking the FCN [44] model on GID dataset is implemented. The quantitative indexes shown in Table V indicate that the selection of the backbone depth has a slight impact on the final

TABLE IV

QUANTITATIVE COMPARISON FOR BASIC UNITS WITH AND WITHOUT TWO CONSTRAINT FACTORS IN MODULE $\boldsymbol{B}$, WHERE THEY ARE ABBREVIATED AS $\boldsymbol{B}$-W/O AND $\boldsymbol{B}$-W, RESPECTIVELY (UNIT: %)

Indexes	$\boldsymbol{B}$-W/O	$\boldsymbol{B}$-W
Water	52.3	81.0
Built-up	74.1	97.6
Farmland	81.8	95.3
Forest	92.4	97.0
Meadow	73.5	85.7
OA	80.1	96.3
Kappa	71.3	93.4

results and leads to the accuracy fluctuation, and ResNet50 can provide the highest numerical result. For the sake of fairness, the depth of ResNet is set to 50 in the following experiments. Moreover, the GSL-CNN method also outperforms other three competition methods employed in Section III-C at different model depths.

C. Semantic Segmentation Experiments

The proposed GSL-CNN method provides a general framework from the sampling perspective that can improve the generalization ability of the CNN-based method by generating training images from raw images using the CBR sampling strategy. To evaluate the effectiveness of the GSL-CNN method, the proposed method is, respectively, tested on GID dataset and Sentinel-2 dataset, whose results are compared with other state-of-the-art data augmentation methods. They are.

1) *Baseline:* It uses two data augmentation tricks, random cropping, and flipping, to generate the training image for the selected CNN method.
2) *Gridmask [30]:* It further uses the regional dropout strategy to remove regional patches from the training image, as illustrated in Fig. 1(b).
3) *Cutmix [32]:* It fills the removed regions with patches from other training images, which can integrate data information from different training images.

To test the effectiveness of the proposed method against different CNN methods, this section also employs 15 state-of-the-art CNN methods. Some of these methods are proposed for general image semantic segmentation, which are FCN, PSPNet [45], PSANet [46], UperNet [47], EncNet [48], DaNet [49], ApcNet [50], CCNet [51], Ann [52], GcNet [53], DnlNet [54], and PointRend [55], respectively. Others are developed for semantic segmentation of the remote sensing image, such as Deeplabv3+ [56], BANet [57], and MANet [58]. By comparing the performances of various CNN methods, it can show the difference between current data augmentation methods and the proposed CBR sampling strategy.

1) Semantic Segmentation on GID Dataset: This section employs 12 different state-of-the-art CNN methods as the CNN module of the GSL-CNN method. They are tested on GID dataset. Under the same experimental conditions, the performances of the above three comparative data augmentation methods are also measured. The numerical results are illustrated in Table VI.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT METHODS AGAINST DIFFERENT DEPTHS

Method	Backbone	Baseline			Gridmask [30]			Cutmix [32]			GSL-CNN		
		OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA
FCN	ResNet18	82.9	70.2	73.0	82.6	71.2	72.3	85.0	71.8	73.7	**86.9**	**77.3**	**76.3**
	ResNet50	82.6	69.3	71.8	82.7	71.0	71.5	85.0	74.6	74.5	**89.6**	**77.5**	**76.2**
	ResNet101	80.6	66.4	69.1	81.3	68.8	70.6	81.6	70.2	70.8	**83.8**	**71.5**	**73.2**

TABLE VI
QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART NETWORKS ON THE GID DATASET TO VERIFY METHOD GENERALIZATION (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Method	Backbone	Baseline			Gridmask [30]			Cutmix [32]			GSL-CNN		
		OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA
FCN [44]	ResNet50	82.6	69.3	71.8	82.7	71.0	71.5	85.0	74.6	74.5	**89.6**	**77.5**	**76.2**
PSPNet [45]	ResNet50	83.6	69.9	72.4	83.4	70.1	72.4	84.7	70.6	74.1	**86.3**	**74.3**	**75.6**
PSANet [46]	ResNet50	77.0	60.3	64.2	83.7	69.7	73.0	84.5	71.0	73.7	**88.0**	**75.7**	**77.1**
EncNet [48]	ResNet50	84.9	71.9	72.8	83.4	70.1	72.9	83.2	71.4	70.8	**86.8**	**77.0**	**77.2**
DaNet [49]	ResNet50	82.0	69.3	71.6	86.3	75.3	73.8	87.5	75.5	75.6	**88.1**	**75.8**	**76.2**
ApcNet [50]	ResNet50	81.6	69.5	71.2	82.7	68.9	69.6	85.5	71.3	73.7	**88.0**	**75.5**	**77.4**
CCNet [51]	ResNet50	81.5	69.6	71.1	82.1	69.4	71.1	83.5	71.6	72.3	**84.2**	**74.9**	**74.5**
Ann [52]	ResNet50	79.4	60.3	64.3	82.4	67.3	68.2	82.5	66.4	68.9	**85.0**	**72.8**	**73.8**
GcNet [53]	ResNet50	79.8	66.8	68.6	79.3	66.5	68.5	85.1	73.5	75.2	**85.7**	**74.3**	**75.8**
DnlNet [54]	ResNet50	82.5	69.9	71.9	82.1	68.9	71.5	83.5	70.9	72.6	**86.3**	**74.1**	**76.5**
BANet [57]	/	87.5	75.6	78.8	88.2	77.2	79.7	88.3	77.6	80.0	**92.4**	**79.9**	**80.7**
MANet [58]	/	85.0	70.4	72.5	87.1	72.9	73.3	87.8	75.2	75.9	**91.1**	**78.1**	**78.6**

As we can observe, the numerical results of 12 different GSL-CNN methods can consistently provide better performances than the baseline method, where the OA indicator can be improved 5.4% and Kappa indicator can be improved 5.7% on average. The GSL-CNN method also outperforms Cutmix and Gridmask in both OA and Kappa indicator, with at least 2.6% and 2.7% indicator improvement on average, respectivaly. To illustrate the advantage of the proposed method in detail, segmentation results of different methods for the DaNet are shown in Fig. 12 as an instance, and their quantitative indexes are given in Table VII. At the first experiment of Fig. 12, built-ups located at the bottom area show different and complex appearances from other areas. It leads to that the result of the baseline can just identify the built-ups in the top area appropriately but fail to recognize the built-ups in the bottom area. Gridmask and Cutmix provide a better performance in the bottom area by introducing the regional augmentation strategy, but there are still some misclassifications. Similar to the first experiment, the appearances of built-ups marked yellow boxes are different from others in the second experiment. It makes the baseline, Gridmask, and Cutmix fail to recognize the built-ups effectively. For the GSL-CNN method, as the CBR strategy helps the model to learn the generalization information of build-up, it can identify build-ups in the above two areas more effectively than others, whose $F1$-score can reach 95.1% and 93.6%, respectively. In the third experiment, the similar spectral responses between water and farmland also make semantic segmentation challenging, such as the area marked yellow box. The GSL-CNN method can provide an appropriate result at these areas as well, where the $F1$-score of water can be improved by 17.8%. In the last experiment, the spatial distribution of water is uneven. It makes the baseline

TABLE VII
QUANTITATIVE RESULTS OF SEMANTIC SEGMENTATION RESULTS ON THE GID TEST IMAGES (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Method	Indexes	Baseline	Gridmask	Cutmix	GSL-CNN
GID_1	Water	97.0	97.6	97.6	96.5
	Built-up	73.9	90.2	85.3	**95.1**
	Farmland	35.4	49.3	42.1	**57.7**
	Forest	**82.8**	78.8	73.4	72.8
	OA	78.8	88.0	84.1	**90.7**
	Kappa	71.9	82.8	78.0	**86.0**
GID_2	Water	96.4	96.8	**97.2**	97.0
	Built-up	62.2	88.7	85.8	**93.6**
	Farmland	55.9	75.7	72.5	**81.9**
	Forest	64.8	59.0	52.8	**70.6**
	OA	66.5	86.3	83.9	**91.3**
	Kappa	60.7	80.5	77.7	**86.4**
GID_3	Water	68.9	64.8	64.9	**86.7**
	Built-up	75.0	78.5	80.3	**82.7**
	Farmland	89.3	89.1	89.4	**93.4**
	Forest	/	/	/	/
	OA	83.9	83.3	84.0	**90.1**
	Kappa	70.2	69.3	70.1	**82.0**
GID_4	Water	72.0	77.9	91.0	**96.3**
	Built-up	75.6	**89.7**	88.8	81.4
	Farmland	79.0	82.6	90.3	**96.4**
	Forest	71.3	70.0	58.8	**75.9**
	OA	76.1	81.0	90.3	**95.2**
	Kappa	67.0	72.9	84.7	**92.0**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

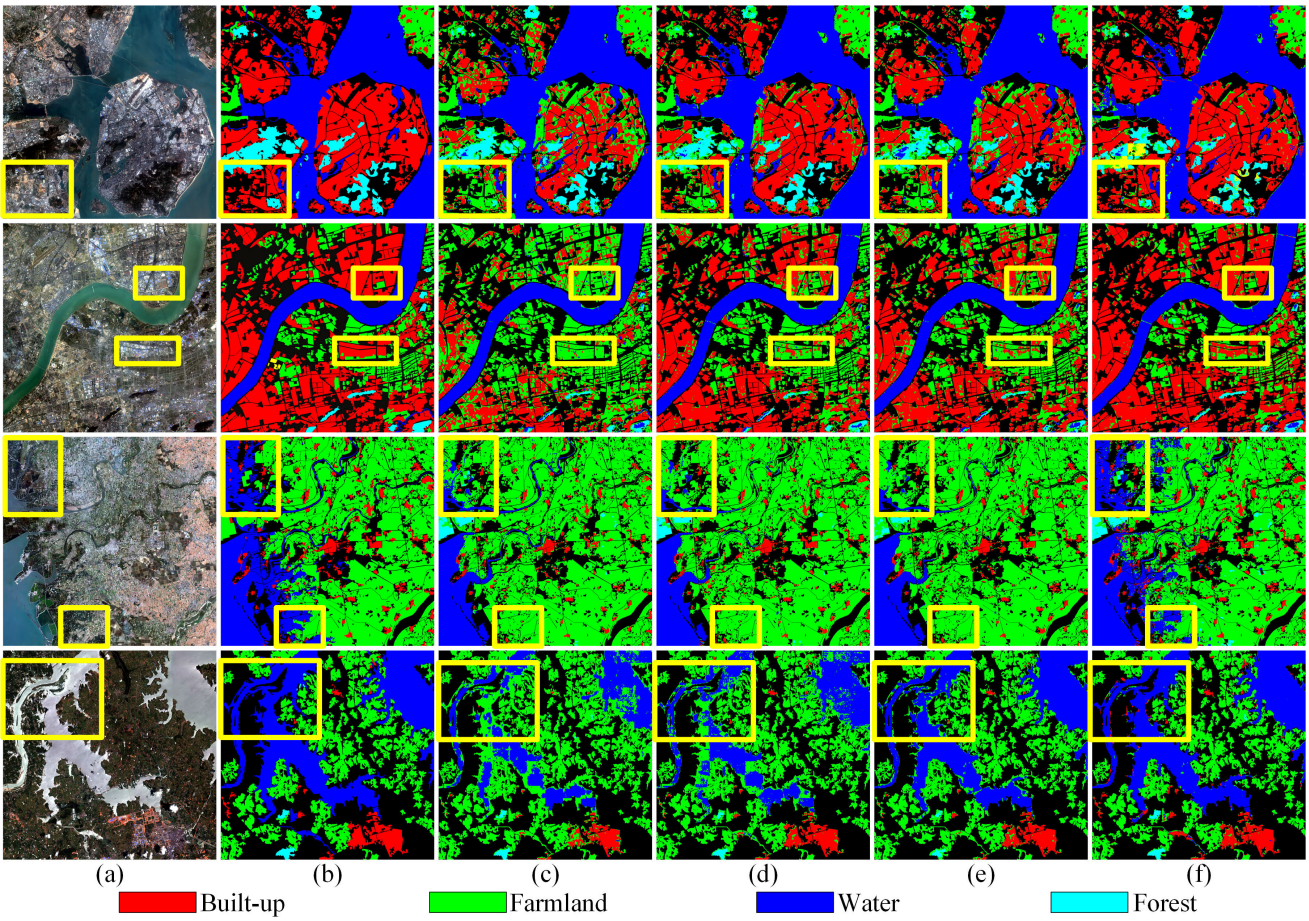


Fig. 12. Semantic segmentation results on GID test images. (a) Original remote sensing images. (b) Ground truth. Results of the FCN with (c) baseline, (d) Gridmask, (e) Cutmix, and (f) GSL-CNN.

and Gridmask fail to recognize the water effectively. The Cutmix can improve the performance but cannot accurately capture the edges of the water. By emphasizing the generalized information, especially with the water-extraction trick, the GSL-CNN method can identify water with different distributions and shapes appropriately, which can achieve higher accuracy, as shown in Table VII.

To qualitatively analyze the generalization ability of the proposed method, gradient-weighted class activation mapping (Grad-CAM) [59] is employed to illustrate the attention through the activation maps of final segmentation, as shown in Fig. 13. Fig. 13(b) shows that the baseline performs poorly in categories except the water. Compared to the baseline, the Gridmask and the Cutmix can capture more discriminative features with the regional augmentation tricks, but these two methods still fail to locate some categories regions effectively, such as built-up and water. Activation maps of these methods demonstrate that interactions between different categories, such as specific spatial distribution, could affect the learning of generalization information and further restrict the performance of models. However, by reducing the attention on the interclass context, the GSL-CNN method can capture the discriminative features of individual category more effectively, as shown in Fig. 13(e). All the above experiments verify that the CBR strategy of the GSL-CNN method can effectively represent the generalization information of different land categories and strengthen the generalization ability of a CNN method.

2) Semantic Segmentation on Sentinel-2 Dataset: To further test the performance of the GSL-CNN method against different datasets, the Sentinel-2 dataset was employed to evaluate the proposed method in this section. Twelve different CNN methods are used as the CNN module of the GSL-CNN method. Their numerical results are shown in Table VIII. Compared with the best results among other three comparison methods, the GSL-CNN method can achieve a maximum improvement of 3.8% on the OA indicator, with an average improvement of 1.8%. The Kappa indicator can be improved 2.9% at the highest and 1.5% on average. These quantitative indexes validate the effectiveness and robustness of the proposed method with different CNN models.

For a more comprehensive evaluation of the GSL-CNN method, results of a specific CNN model, the BANet, are illustrated in Fig. 14, and numerical results of individual categories are shown in Table IX. In the first three experiments, there are many similar appearances between different categories, such as farmland and water in the first experiment, forest and water in the third experiment. It makes many misclassifications in the results of the baseline and Gridmask method. The Cutmix can help CNN method learn originally less sensitive and important information through regional replacement. Hence, compared to baseline and Gridmask, it can provide a better performance, such as the identification of water in the third experiment. But, if the appearances of different land categories are extremely similar, Cutmix will lead to unfavorable results,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

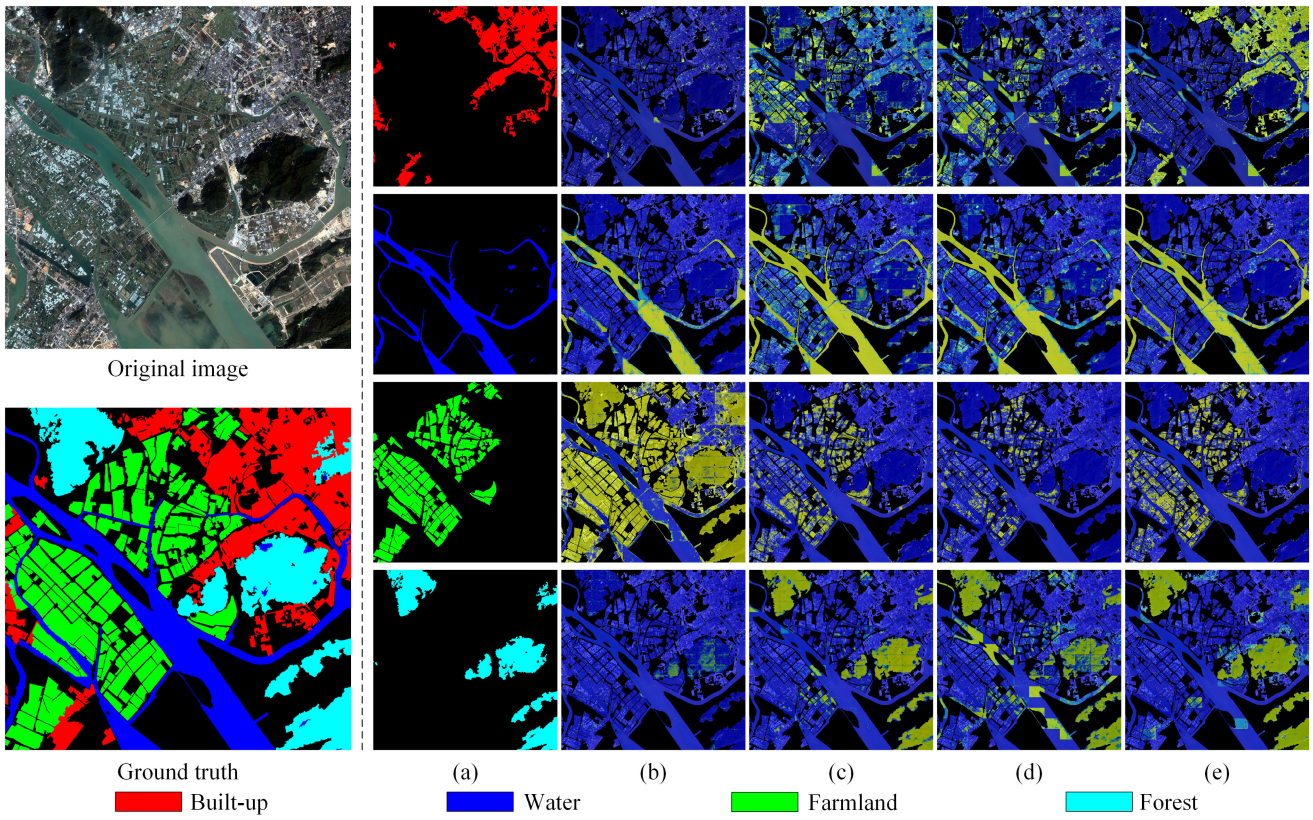


Fig. 13. Activation maps of different methods on the GID dataset. (a) Ground truth of each category. (b) Baseline. (c) Gridmask. (d) Cutmix. (e) GSL-CNN.

TABLE VIII

QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART NETWORKS ON SENTINEL-2 DATASET TO VERIFY METHOD GENERALIZATION (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Method	Backbone	Baseline			Gridmask [30]			Cutmix [32]			GSL-CNN		
		OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA
FCN [44]	ResNet50	80.2	73.7	69.1	86.7	77.6	76.6	88.1	81.2	78.5	**89.6**	**82.6**	**80.6**
PSPNet [45]	ResNet50	80.2	70.3	68.9	87.5	78.8	77.3	86.8	80.8	77.3	**88.9**	**81.1**	**78.8**
Deeplabv3+ [56]	ResNet50	75.6	71.2	64.6	87.3	81.4	77.4	87.9	81.5	78.6	**90.6**	**82.1**	**80.8**
PSANet [46]	ResNet50	76.3	77.5	64.9	86.8	78.6	76.6	86.9	80.6	**77.3**	**87.5**	**80.8**	77.1
UperNet [47]	ResNet50	78.2	73.9	67.1	85.6	77.8	75.8	87.5	80.0	77.9	**89.3**	**81.1**	**79.0**
DaNet [49]	ResNet50	85.9	75.2	74.4	87.9	79.7	78.3	87.7	82.4	79.3	**89.2**	**82.5**	**79.4**
ApcNet [50]	ResNet50	75.7	66.7	64.8	76.6	69.2	65.1	85.0	**80.2**	**76.1**	**86.4**	79.9	75.7
GcNet [53]	ResNet50	81.0	76.2	70.6	87.8	79.4	77.3	87.7	81.5	77.9	**88.9**	**82.3**	**79.3**
DnlNet [54]	ResNet50	74.4	66.6	62.7	86.9	78.1	76.9	87.5	81.3	77.9	**90.0**	**82.8**	**79.9**
PonitRend [55]	ResNet50	76.7	73.8	66.4	82.1	76.9	72.1	86.9	81.0	78.0	**89.1**	**82.7**	**78.4**
BANet [57]	/	86.9	79.5	75.2	87.7	79.9	77.2	89.0	82.1	78.8	**90.5**	**83.2**	**80.0**
MANet [58]	/	86.6	81.9	77.2	87.8	82.0	79.7	88.0	82.6	78.2	**91.8**	**83.9**	**82.6**

such as water in the first experiment and farmland in the second experiment. By distinguishing the difference between the intraclass and the interclass contexts with the CBR strategy, the GSL-CNN method can effectively reduce these misclassifications and provide a better performance consistently, and its numerical results further validate the superiority of the GSL-CNN method. The last experiment shows a special case for semantic segmentation, where forest covers almost all regions while other land categories are only sparsely distributed in this test image. The data distribution of this experiment is quite different from other specific data distributions, especially the spatial distribution of built-up and farmland. It causes that three comparison methods fail to effectively learn the forest information and wrongly label some forest areas as built-up or farmland, whose numerical result of the built-up and the farmland are also quite limited. However, training images generated by the CBR strategy help the GSL-CNN method focus on learning the generalization information of the forest and reducing the impact of the interclass context. It makes a better performance, where the $F1$-score can be increased at least 25.2% for built-up, 12.3% for farmland, and 7.5% for forest. Above experiments further verify that the GSL-CNN

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

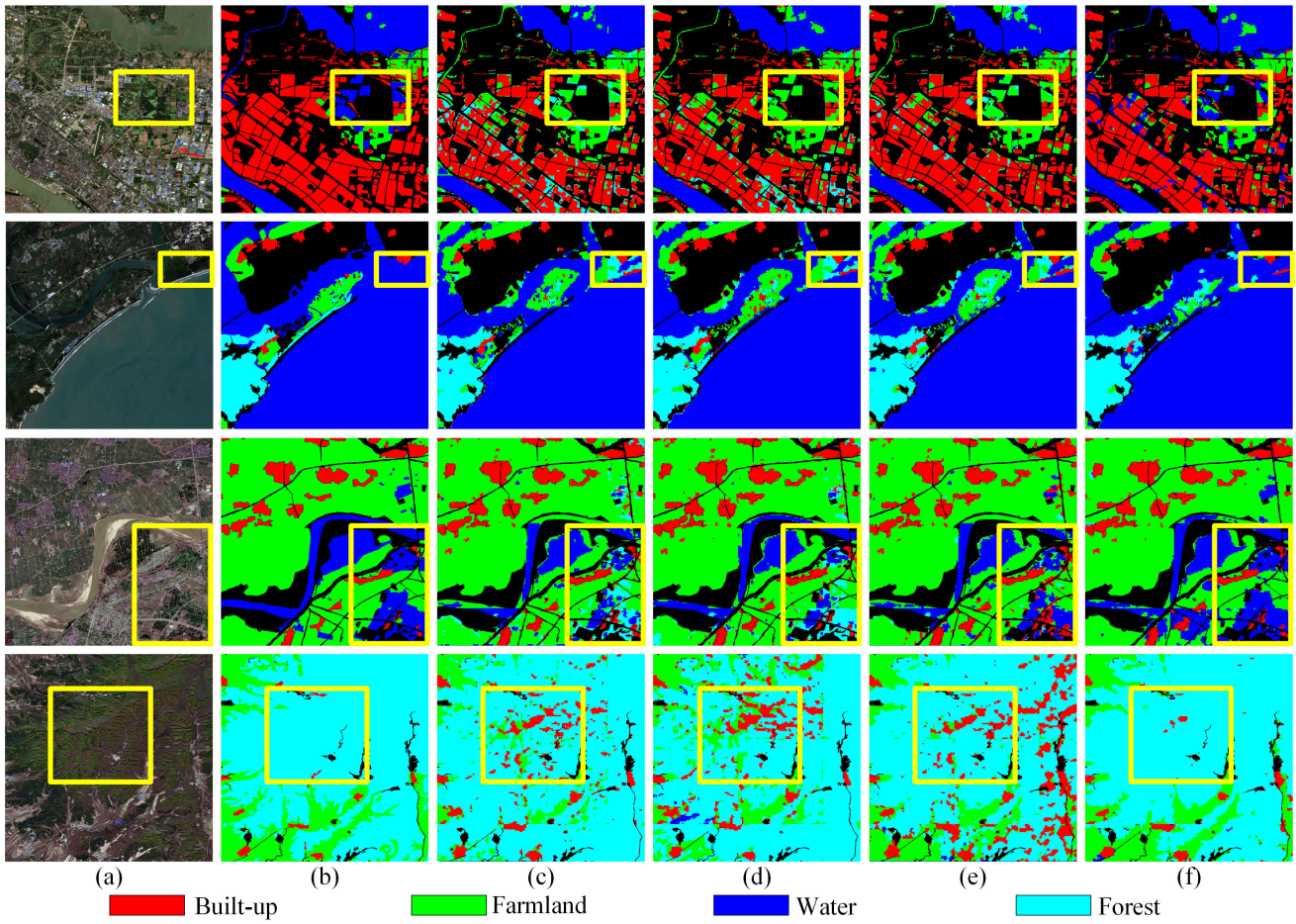


Fig. 14. Semantic segmentation results on the Sentinel-2 test images. (a) Original remote sensing images. (b) Ground truth. Results of the BANet with (c) baseline, (d) Gridmask, (e) Cutmix, and (f) GSL-CNN.

TABLE IX
QUANTITATIVE RESULTS OF SEMANTIC SEGMENTATION RESULTS ON THE SENTINEL-2 TEST IMAGES (UNIT: %). THE HIGHEST VALUES OF EVERY METRIC ARE IN BOLD

Method	Indexes	Baseline	Gridmask	Cutmix	GSL-CNN
GEE_1	Water	87.0	87.0	86.3	**88.4**
	Built-up	87.3	87.8	**93.0**	91.6
	Farmland	49.0	46.1	50.8	**61.8**
	Forest	/	/	/	/
	OA	78.4	79.3	84.0	**87.3**
	Kappa	69.8	70.7	75.7	**80.4**
GEE_2	Water	94.1	94.8	94.5	**96.3**
	Built-up	81.5	75.6	79.0	**82.3**
	Farmland	57.8	**62.9**	56.6	60.2
	Forest	82.9	81.9	79.2	**83.2**
	OA	89.3	90.1	89.0	**92.1**
	Kappa	75.9	77.4	75.6	**79.6**
GEE_3	Water	69.1	73.0	77.7	**81.4**
	Built-up	81.0	79.3	76.2	**84.8**
	Farmland	92.4	93.4	**93.5**	93.4
	Forest	/	/	/	/
	OA	85.8	86.7	88.8	**90.2**
	Kappa	73.2	74.7	78.2	**81.4**
GEE_4	Water	/	/	/	/
	Built-up	30.8	26.6	18.2	**56.0**
	Farmland	39.9	48.3	67.9	**80.2**
	Forest	87.2	84.8	87.9	**95.4**
	OA	77.6	74.9	79.2	**91.5**
	Kappa	54.6	55.2	62.1	**79.4**

method can robustly enhance the generalization ability and improve the segmentation accuracy for different CNN models.

IV. CONCLUSION

A GSL-CNN method is proposed in this article to enhance the generalization ability of the CNN-based model for semantic segmentation of the remote sensing image. The strength of the proposed method mainly lies in two aspects. One aspect is that a new CBR sampling strategy is developed to emphasize the generalization information learning of each category from the perspective of sampling when generating the training images from the raw large-size annotated image. The other aspect is that the GSL-CNN method provides a framework to combine the CBR sampling strategy with the CNN-based method for semantic segmentation.

The influence of the CBR sampling strategy in the GSL-CNN method is thoroughly discussed, showing that the category grouping module ($\boldsymbol{C}$) and the basic unit extraction module ($\boldsymbol{B}$) can focus on the representation of the generalization information of each category and the exploration of the intraclass contextual information, and the random combination module ($\boldsymbol{R}$) can reduce the impact of the interclass contextual information by randomly generating different data distributions. The proposed method, integrating the CBR sampling strategy with different CNN methods, is experimentally

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

demonstrated to have the potential of enhancing the generalization ability of the CNN-based methods and improving the accuracy of the semantic segmentation. The effectiveness and robustness are further tested on various remote sensing images with 15 different CNN methods, showing that the CNN methods coupled with the CBR strategy, i.e., the GSL-CNN method with different CNN modules, can provide better performance than these methods without the CBR strategy. The future work will be focused on further improving the CBR sampling strategy from two aspects: 1) developing basic units with adaptive shapes to more effectively capture the generalization concept of different land categories and 2) integrating the geoscience laws into the random combination module ($\boldsymbol{R}$) to guide the generation of data distribution.

ACKNOWLEDGMENT

The authors are very grateful to the editors and the anonymous referees for their constructive and valuable comments and suggestions, which led to the present greatly improved version of their manuscript. Especially, the introduction and discussion of the CBR sampling strategy and the role of each module are improved according to the anonymous referees.

REFERENCES

[1] M. Wurm, T. Stark, X. X. Zhu, M. Weigand, and H. Taubenböck, "Semantic segmentation of slums in satellite images using transfer learning on fully convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 59–69, Apr. 2019.

[2] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-a: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[3] K. Fu, Z. Chang, Y. Zhang, G. Xu, K. Zhang, and X. Sun, "Rotation-aware and multi-scale convolutional neural network for object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 294–308, Mar. 2020.

[4] Z. S. Venter, O. Brousse, I. Esau, and F. Meier, "Hyperlocal mapping of urban air temperature using remote sensing and crowdsourced weather data," *Remote Sens. Environ.*, vol. 242, Jun. 2020, Art. no. 111791.

[5] C. Xu, C. Li, Z. Cui, T. Zhang, and J. Yang, "Hierarchical semantic propagation for object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4353–4364, Jun. 2020.

[6] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[7] Y. Shen et al., "Efficient deep learning of nonlocal features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6029–6043, Jul. 2021.

[8] Q. Zhu, C. Liao, H. Hu, X. Mei, and H. Li, "MAP-Net: Multiple attending path neural network for building footprint extraction from remote sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6169–6181, Jul. 2021.

[9] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603018.

[10] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[11] Z. Liu, L. Ma, and Q. Du, "Class-wise distribution adaptation for unsupervised classification of hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 508–521, Jan. 2021.

[12] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 675–685, Jan. 2021.

[13] Z. Fang, G. Zhang, Q. Dai, Y. Kong, and P. Wang, "Semisupervised deep convolutional neural networks using pseudo labels for PolSAR image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] B. Hobley, R. Arosio, G. French, J. Bremner, T. Dolphin, and M. Mackiewicz, "Semi-supervised segmentation for coastal monitoring seagrass using RPA imagery," *Remote Sens.*, vol. 13, no. 9, p. 1741, Apr. 2021.

[15] L. Zhang and J. Ma, "Salient object detection based on progressively supervised learning for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9682–9696, Nov. 2021.

[16] Y. Lin, G. Vosselman, and M. Y. Yang, "Weakly supervised semantic segmentation of airborne laser scanning point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 79–100, May 2022.

[17] D. Hong, N. Yokoya, N. Ge, J. Chanussot, and X. X. Zhu, "Learnable manifold alignment (LeMA): A semi-supervised cross-modality learning framework for land cover and land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 193–205, Jan. 2019.

[18] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[19] S. Wang, W. Chen, S. M. Xie, G. Azzari, and D. B. Lobell, "Weakly supervised deep learning for segmentation of remote sensing imagery," *Remote Sens.*, vol. 12, no. 2, p. 207, Jan. 2020.

[20] X. Kang, B. Zhuo, and P. Duan, "Semi-supervised deep learning for hyperspectral image classification," *Remote Sens. Lett.*, vol. 10, no. 4, pp. 353–362, Apr. 2019.

[21] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[22] C. Wen, X. Sun, J. Li, C. Wang, Y. Guo, and A. Habib, "A deep learning framework for road marking extraction, classification and completion from mobile laser scanning point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 178–192, Jan. 2019.

[23] S. Ji, D. Wang, and M. Luo, "Generative adversarial network-based full-space domain adaptation for land cover classification from multiple-source remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 3816–3828, May 2021.

[24] Y. Gong et al., "Enlighten-GAN for super resolution reconstruction in mid-resolution remote sensing images," *Remote Sens.*, vol. 13, no. 6, p. 1104, Mar. 2021.

[25] N. Lv et al., "Remote sensing data augmentation through adversarial training," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9318–9333, 2021.

[26] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[27] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015, pp. 1–14.

[28] A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 2, 2012, pp. 1097–1105.

[29] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *J. Big Data*, vol. 6, no. 1, pp. 1–48, Jul. 2019.

[30] P. Chen, S. Liu, H. Zhao, and J. Jia, "GridMask data augmentation," 2020, *arXiv:2001.04086*.

[31] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," 2017, *arXiv:1710.09412*.

[32] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6022–6031.

[33] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[34] G. J. Scott, M. R. England, W. A. Starms, R. A. Marcum, and C. H. Davis, "Training deep convolutional neural networks for land-cover classification of high-resolution imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 549–553, Apr. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.

[35] Y. Zhong, Z. Zheng, A. Ma, X. Lu, and L. Zhang, "COLOR: Cycling, offline learning, and online representation framework for airport and airplane detection using GF-2 satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8438–8449, Dec. 2020.

[36] Y. Shen et al., "BDANet: Multiscale convolutional neural network with cross-directional attention for building damage assessment from satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5402114.

[37] W. G. Cochran, *Sampling Techniques*, 3rd ed. Hoboken, NJ, USA: Wiley, 1977.

[38] X. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[39] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sens. Environ.*, vol. 202, pp. 18–27, Dec. 2017.

[40] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014, pp. 1–15.

[41] X. Zheng, L. Huan, G.-S. Xia, and J. Gong, "Parsing very high resolution urban scene images by learning deep ConvNets with edge-aware loss," *ISPRS J. Photogramm. Remote Sens.*, vol. 170, pp. 15–28, Dec. 2020.

[42] L. Mou, S. Saha, Y. Hua, F. Bovolo, L. Bruzzone, and X. X. Zhu, "Deep reinforcement learning for band selection in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504414.

[43] F. Mohammadimanesh, B. Salehi, M. Mahdianpari, E. Gill, and M. Molinier, "A new fully convolutional neural network for semantic segmentation of polarimetric SAR imagery in complex land cover ecosystem," *ISPRS J. Photogramm. Remote Sens.*, vol. 151, pp. 223–236, May 2019.

[44] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[45] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[46] H. Zhao et al., "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 270–286.

[47] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and S. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 432–448.

[48] H. Zhang et al., "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160.

[49] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.

[50] J. He, Z. Deng, L. Zhou, Y. Wang, and Y. Qiao, "Adaptive pyramid context network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7511–7520.

[51] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[52] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 593–602.

[53] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980.

[54] M. Yin et al., "Disentangled non-local neural networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Nov. 2020, pp. 191–207.

[55] A. Kirillov, Y. Wu, K. He, and R. Girshick, "PointRend: Image segmentation as rendering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9796–9805.

[56] L. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2018, pp. 801–818.

[57] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021.

[58] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[59] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



Chen Zheng (Member, IEEE) received the B.S. degree from the School of Mathematics and Information Sciences, Henan University, Kaifeng, Henan, China, in 2007, and the M.S. and Ph.D. degrees from the School of Mathematics and Statistics, Wuhan University, Wuhan, Hubei, China, in 2009 and 2012, respectively.

He did his post-doctoral research at the National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China. He was a Post-Doctoral Researcher with the University of New Brunswick, Fredericton, NB, Canada. Since 2023, he has been a Professor with the School of Mathematics and Statistics, Henan University. He was in-charge of a research project on semantic image segmentation from the National Natural Science Foundation of China. He is the author of 29 articles. His research interests include remote sensing image segmentation and multilevel semantic analysis of remote sensing images.



Jingying Li (Graduate Student Member, IEEE) received the B.S. degree from the School of Mathematical Sciences, Dezhou University, Dezhou, China, in 2018, and the M.S. degree from the School of Mathematics and Statistics, Henan University, Kaifeng, China, in 2021, where she is currently pursuing the Ph.D. degree.

Her research interests include remote sensing image segmentation and deep learning.



Yuncheng Chen (Graduate Student Member, IEEE) received the B.S. and M.S. degrees from the School of Mathematics and Statistics, Henan University, Kaifeng, China, in 2017 and 2020, respectively, where she is currently pursuing the Ph.D. degree.

Her research interests include remote sensing image segmentation, deep learning, and multilevel semantic analysis of remote sensing images.



Leiguang Wang received the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, Hubei, China, in 2009.

From 2014 to 2015, he was a Post-Doctoral Researcher with the University of New Brunswick, Fredericton, NB, Canada. He is currently a Professor with Southwest Forestry University, Kunming, China, where he has been the Vice Dean of the Institute of Big Data and Artificial Intelligence since 2018. He is the author of more than 50 articles. His research interests include remote sensing image fusion, semantic segmentation, and application in forestry.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:02:32 UTC from IEEE Xplore. Restrictions apply.